RootCause AI - Backend Master Prompt (Part 1)

Act as a Senior Backend Engineer and AI Engineer.

Build a production-ready backend for RootCause AI.

Tech Stack:
- Node.js
- Express.js
- MongoDB Atlas + Mongoose
- JWT Authentication
- bcrypt
- Multer
- Google Gemini 2.5 Flash
- Helmet
- CORS
- express-rate-limit
- cookie-parser
- express-validator
- dotenv
- Morgan

Architecture:
src/
config/
controllers/
middleware/
models/
routes/
services/
prompt/
validators/
utils/
uploads/
app.js
server.js

Authentication:
- Register
- Login
- Logout
- JWT in HTTP-only cookies
- bcrypt password hashing
- Protected routes
- Refresh-token ready architecture

MongoDB Models:
User:
name,email,password,createdAt,updatedAt

Analysis:
userId,bugTitle,bugReport,consoleLogs,stackTrace,
codeSnippet,environment,screenshot,uploadedFiles,
analysis(JSON),severity,priority,confidence,
createdAt,updatedAt

Application Flow:
Landing -> Login -> Dashboard -> New Analysis
Upload Files -> POST /api/analysis/upload
Store files -> Extract text -> Build Prompt
Call Gemini API -> Receive JSON
Store in MongoDB
Return:
{
success:true,
analysisId:"..."
}
Frontend redirects to:
/app/history/:analysisId
Then frontend requests:
GET /api/analysis/:analysisId

AI JSON:
summary
reproductionSteps
rootCause
confidence
suggestedFiles
fixRecommendations
severity
priority

Security:
Helmet
Rate Limiter
Input Validation
HTTP-only Cookies
CORS
File Validation
Secure Error Handling
MongoDB Injection Protection
XSS Protection

Environment Variables:
PORT
MONGODB_URI
JWT_SECRET
JWT_EXPIRE
COOKIE_EXPIRE
GEMINI_API_KEY
NODE_ENV

Note:
This PDF is Part 1 because the complete master prompt exceeds a single document response in chat. Continue by expanding controllers, services, middleware, routes, validators, and promptBuilder using this architecture.